

Inverse Isotope Effect: A Brief Interpretation Based on Anharmonic Effects in Lattice Vibrations and Electron-Phonon Coupling

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Abstract

The inverse isotope effect observed in palladium hydrides (PdH_x) presents a significant challenge to the BCS theory, where the superconducting critical temperature (T_c) is expected to scale with the isotopic mass as $M^{-0.5}$. In this report, I interpret this anomalous phenomenon by incorporating anharmonic effects into lattice vibrations. First, I approximately calculate the electron-phonon coupling constant (λ) starting from the Fröhlich Hamiltonian. Subsequently, using a one-dimensional anharmonic toy model, I demonstrate the significant anharmonic corrections to the effective force constant and its impact on electron-phonon coupling constant by the large zero-point motion. Finally, by applying the McMillan equation, I show that strong anharmonicity can reverse the dependence of T_c on mass, providing a qualitative explanation for the inverse isotope effect.

1 Introduction

The microscopic mechanism about superconductivity remains a central topic in condensed matter physics since its discovery in 1911 in mercury (Hg). Although the classical Drude-Sommerfeld model successfully explains how electrons conduct electricity in metals, it fails to account for the phenomenon of superconductivity, where certain materials exhibit zero electrical resistance below a critical temperature (T_c).

The breakthrough in understanding superconductivity came in 1957 with the Bardeen-Cooper-Schrieffer (BCS) theory[1]. The BCS theory posits that an attractive interaction between electrons, caused by electron and lattice vibrations (phonons) coupling, leads to the formation of Cooper pairs. And the isotope effect[6] is one of the critical experimental test of this electron-phonon coupling mechanism, which relies on the relationship between ionic mass and vibration frequency. In the interpretation of isotope effect, we assume that the phonon frequencies are harmonic. Then the superconducting critical temperature T_c will behave as $T_c \propto M^{-\alpha}$, where M is the ionic mass and $\alpha = 0.5$ is the isotope coefficient.

However, in many superconducting materials like MgB_2 [4], the isotope coefficient is substantially reduced. And in the most extreme case of palladium hydrides, the isotope effect is inverted, with α taking negative values[10]. This inverse isotope effect challenges our theories. Actually, take

palladium hydrides as an example, PdH has $T_c = 8 - 9K$. Meanwhile, hydrogen substitution with the heavier deuterium leads to a higher T_c , i.e., $T_c = 10 - 11K$ for PdD[2]. Because the ratio of the mass of deuterium to hydrogen is about 2, the isotope coefficient α can be calculated as:

$$\alpha \simeq -\frac{\ln T_c(PdD) - \ln T_c(PdH)}{\ln 2} \simeq -0.3 < 0 \quad (1)$$

This anomalous phenomenon indicates that the harmonic approximation fails for light elements like hydrogen, where the large zero-point motion of ions reaches the anharmonic regions of the lattice potential. Therefore, to understand the inverse isotope effect, we need to consider the anharmonic effects in lattice vibrations and its influence on electron-phonon coupling.

In this article, I will first briefly calculate the electron-phonon coupling constant λ from the Fröhlich Hamiltonian. Subsequently, I employ a one-dimensional anharmonic potential toy model to analyze the impact of anharmonic effects on this coupling. Finally, by applying the McMillan equation, I analyze the superconducting critical temperature T_c and provide a brief explanation for the inverse isotope effect.

2 Brief interpretation of Inverse Isotope Effect

2.1 Fröhlich Hamiltonian and Electron-Phonon Coupling Constant

First, we can write the periodic lattice potential as:

$$V_{tot}(\mathbf{r}) = \sum_l V_{ion}(\mathbf{r} - \mathbf{R}_l^0 - u_l) \quad (2)$$

where V_{ion} is the potential from a single ion, $\mathbf{R}_l^0$ is the equilibrium position of the l -th ion, and u_l is its displacement. Expanding $V_{tot}(\mathbf{r})$ to the first order in u_l , we have:

$$V_{tot}(\mathbf{r}) \simeq \sum_l [V_{ion}(\mathbf{r} - \mathbf{R}_l^0) - u_l \cdot \nabla V_{ion}(\mathbf{r} - \mathbf{R}_l^0)] \quad (3)$$

The first term represents the static lattice potential, while the second term describes the electron-phonon interaction, i.e., H_{e-ph} .

We can express the ionic displacement u_l in terms of phonon creation and annihilation operators with wavevector $\mathbf{q}$ and mode ν [5]:

$$u_l = \sum_{\mathbf{q}\nu} \sqrt{\frac{\hbar}{2NM\omega_{\mathbf{q}\nu}}} \hat{e}_{\mathbf{q}\nu} \left(b_{\mathbf{q}\nu} e^{i\mathbf{q}\cdot\mathbf{R}_l^0} + b_{\mathbf{q}\nu}^\dagger e^{-i\mathbf{q}\cdot\mathbf{R}_l^0} \right) \quad (4)$$

where M is the ionic mass, $\omega_{\mathbf{q}\nu}$ is the phonon frequency, and $\hat{e}_{\mathbf{q}\nu}$ is the polarization vector. N is the number of unit cells in the crystal.

Then use the second quantization formalism, we can write the electron-phonon interaction Hamil-

tonian as Fröhlich Hamiltonian[3]:

$$H_{e-ph} = \sum_{\mathbf{k}\mu\mu'\sigma} \sum_{\mathbf{q}\nu} g_{\mathbf{k}+\mathbf{q}\mu',\mathbf{k}\mu}^{\mathbf{q}\nu} c_{\mathbf{k}+\mathbf{q}\mu'\sigma}^\dagger c_{\mathbf{k}\mu\sigma} (b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger) \quad (5)$$

where $c_{\mathbf{k}\mu\sigma}^\dagger$ and $c_{\mathbf{k}\mu\sigma}$ are the electron creation and annihilation operators with momentum $\mathbf{k}$, band index μ , and spin σ , respectively, and $g_{\mathbf{k}+\mathbf{q}\mu',\mathbf{k}\mu}^{\mathbf{q}\nu}$ is the electron-phonon coupling matrix element given by:

$$g_{\mathbf{k}+\mathbf{q}\mu',\mathbf{k}\mu}^{\mathbf{q}\nu} = \sqrt{\frac{\hbar}{2NM\omega_{\mathbf{q}\nu}}} \langle \mathbf{k} + \mathbf{q}\mu' \sigma | \hat{\mathbf{e}}_{\mathbf{q}\nu} \cdot \nabla V_{ion} | \mathbf{k}\mu\sigma \rangle \quad (6)$$

Then, we can use perturbation theory to calculate the effective interaction matrix element of the electron-phonon coupling. Because of $b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger$, the first-order correction is zero. The second-order correction gives the first non-zero contribution:

$$V_{eff} = \sum_m \frac{\langle f | H_{e-ph} | m \rangle \langle m | H_{e-ph} | i \rangle}{E_i - E_m} \quad (7)$$

For the case of two electrons with momenta $\mathbf{k}$ and $\mathbf{p}$ ($|i\rangle = c_{\mathbf{k}m's'}^\dagger c_{\mathbf{p}n's}^\dagger |0\rangle_e \otimes |0\rangle_{ph}$) scattering to $\mathbf{k} + \mathbf{q}$ and $\mathbf{p} - \mathbf{q}$ ($|f\rangle = c_{\mathbf{k}+\mathbf{q}m's'}^\dagger c_{\mathbf{p}-\mathbf{q}n's}^\dagger |0\rangle_e \otimes |0\rangle_{ph}$), the mediators are:

$$|m_1\rangle = c_{\mathbf{k}+\mathbf{q}m's'}^\dagger c_{\mathbf{p}n's}^\dagger |0\rangle_e \otimes b_{-\mathbf{q}\nu}^\dagger |0\rangle_{ph} \quad \text{or} \quad |m_2\rangle = c_{\mathbf{k}m's'}^\dagger c_{\mathbf{p}-\mathbf{q}n's}^\dagger |0\rangle_e \otimes b_{\mathbf{q}\nu}^\dagger |0\rangle_{ph} \quad (8)$$

Then for a given phonon mode $(\mathbf{q}, \nu)$, the effective interaction matrix element becomes:

$$V_{eff}(\mathbf{k}, \mathbf{p}, \mathbf{q}, \nu) = g_{\mathbf{k}+\mathbf{q}m',\mathbf{k}m}^{\mathbf{q}\nu} \cdot g_{\mathbf{p}-\mathbf{q}n',\mathbf{p}n}^{-\mathbf{q}\nu} \left(\frac{1}{\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}+\mathbf{q}} - \hbar\omega_{-\mathbf{q}\nu}} + \frac{1}{\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}-\mathbf{q}} - \hbar\omega_{\mathbf{q}\nu}} \right) \quad (9)$$

where $\epsilon_{\mathbf{k}}$ is the electron energy with momentum $\mathbf{k}$, $\omega_{\mathbf{q}\nu} = \omega_{-\mathbf{q}\nu}$ is the phonon frequency with momentum $\mathbf{q}$ and mode index ν .

Then let $\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}-\mathbf{q}} = \hbar\Omega = -(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}+\mathbf{q}})$, i.e., the energy transfer during the scattering process, we can rewrite the effective interaction as:

$$V_{eff} = g^2 \cdot \frac{2\hbar\omega_{\mathbf{q}\nu}}{(\hbar\Omega)^2 - (\hbar\omega_{\mathbf{q}\nu})^2} \quad (10)$$

where $g^2 = g_{\mathbf{k}+\mathbf{q}m',\mathbf{k}m}^{\mathbf{q}\nu} \cdot g_{\mathbf{p}-\mathbf{q}n',\mathbf{p}n}^{-\mathbf{q}\nu} \propto \frac{1}{M\omega_{\mathbf{q}\nu}}$

According to [7], the electron-phonon coupling constant λ in the strong coupling theory is corresponding to the $N(0)V$ in the BCS theory, where $N(0)$ is the electronic density of states at the Fermi surface and V is the pairing potential arising from the electron-phonon interaction. And also according to [7], in order to effectively reduce the complex retarded electron-phonon interaction to a constant mass renormalization factor $Z(0) = 1 + \lambda$, we choose $\Omega \ll \omega_{\mathbf{q}\nu}$. So let's approximate λ as:

$$\lambda = N(0) \langle V_{eff} \rangle_{FS} = N(0) \sum_{\mathbf{q}\nu} \langle g^2 \cdot \frac{2\hbar\omega_{\mathbf{q}\nu}}{(\hbar\Omega)^2 - (\hbar\omega_{\mathbf{q}\nu})^2} \rangle_{FS} \simeq -N(0) \sum_{\mathbf{q}\nu} \langle \frac{2g^2}{\hbar\omega_{\mathbf{q}\nu}} \rangle_{FS} \quad (11)$$

where $\langle \dots \rangle_{FS}$ denotes the average over the Fermi surface.

If we consider the lattice vibrations within the harmonic approximation, the phonon frequency $\omega_{\mathbf{q}\nu} \propto M^{-1/2}$. Then:

$$\lambda \propto - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{\frac{M\omega_{\mathbf{q}\nu}}{\omega_{\mathbf{q}\nu}}} \right\rangle_{FS} = - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{M\omega_{\mathbf{q}\nu}^2} \right\rangle_{FS} = - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{k_{\mathbf{q}\nu}} \right\rangle_{FS} \quad (12)$$

which is independent of the ionic mass M .

2.2 One-dimensional Anharmonic Potential Toy Model

To illustrate the impact of anharmonic effects on electron-phonon coupling, we consider a one-dimensional anharmonic potential toy model:

$$V(x) = \frac{1}{2}kx^2 + \alpha x^3 + \beta x^4 (\beta > 0) \quad (13)$$

where k is the harmonic force constant, and α and β are the anharmonic coefficients. Then let's introduce an effective harmonic force constant k_{eff} to approximate the anharmonic potential:

$$k_{eff} = \left\langle \frac{d^2V}{dx^2} \right\rangle_{ground-state(gs)} = k + 6\alpha \langle x \rangle_{gs} + 12\beta \langle x^2 \rangle_{gs} \quad (14)$$

Use the ground state wavefunction of the harmonic oscillator, we have:

$$\langle x \rangle_{gs} = 0, \quad \langle x^2 \rangle_{gs} = \frac{\hbar}{2\sqrt{kM}} \quad (15)$$

From Eq.15 we can see that for light elements, because of the larger zero-point motion, ions will reach the anharmonic regions of the lattice potential more easily.

Then the effective harmonic force constant becomes:

$$k_{eff} = k + \frac{6\beta\hbar}{\sqrt{kM}} \quad (16)$$

Then for a given phonon mode $(\mathbf{q}, \nu)$, the effective phonon frequency considering anharmonic effects is:

$$\omega_{\mathbf{q}\nu} = \sqrt{\frac{k_{eff}}{M}} = \sqrt{\frac{k_{\mathbf{q}\nu}}{M} + \frac{6\beta\hbar}{M\sqrt{k_{\mathbf{q}\nu}M}}} \quad (17)$$

2.3 Critical Temperature and Inverse Isotope Effect

According to Eq.12 and Eq.17, we can get that:

$$\lambda \propto - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{M\omega_{\mathbf{q}\nu}^2} \right\rangle_{FS} = - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{M\left(\frac{k_{\mathbf{q}\nu}}{M} + \frac{6\beta\hbar}{M\sqrt{k_{\mathbf{q}\nu}M}}\right)} \right\rangle_{FS} = - \sum_{\mathbf{q}\nu} \left\langle \frac{1}{k_{\mathbf{q}\nu} + \frac{6\beta\hbar}{\sqrt{k_{\mathbf{q}\nu}M}}} \right\rangle_{FS} \quad (18)$$

According to [9] and [8], we should consider superconductors like palladium hydrides as strong-coupled superconductors. Then let's use McMillan equation[7] to analyze the superconducting critical temperature T_c :

$$T_c = \frac{\hbar\omega_D}{1.45k_B} \exp \left[-\frac{1.04(1+\lambda)}{\lambda - \mu^*(1+0.62\lambda)} \right] \quad (19)$$

where ω_D is the Debye frequency, k_B is the Boltzmann constant, and μ^* is the Coulomb pseudopotential.

Let's choose Debye temperature $T_D = \frac{\hbar\omega_D}{k_B} (\sim 800\text{K}) \propto M^{-1/2}$, and $\mu^* \sim 0.11$ as typical values. And for simplicity, let's set $\lambda \sim \frac{A}{k + \frac{\beta}{\sqrt{M}}}$ and choose the mass of hydrogen $M_H = 1$ as the unit of mass, where A and k are constants related to material properties. And I choose $A = 0.95$, $k = 1.0$ and $\beta = 0.0, 0.8, 1.6$. Then we can plot the superconducting critical temperature $\ln(T_c)$ as a function of ionic mass $\ln(M)$ for different values of anharmonic coefficient β (see Fig.1).

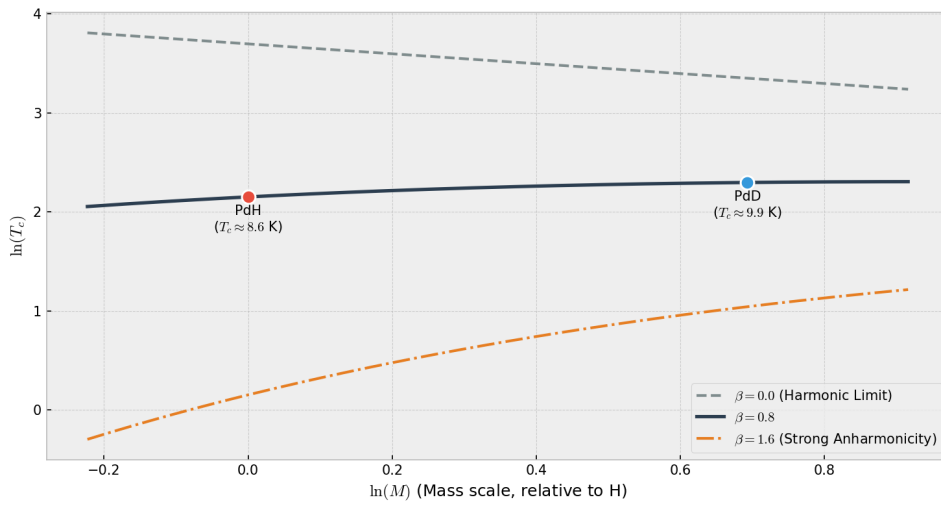


Figure 1: Superconducting critical temperature $\ln(T_c)$ as a function of ionic mass $\ln(M)$ for different values of anharmonic coefficient β . The grey, black, and orange curves correspond to $\beta = 0.0, 0.8, 1.6$, respectively.

From Fig.1, we can see that when the anharmonic coefficient β is small (grey curve), the superconducting critical temperature T_c decreases with increasing ionic mass M , corresponding to the normal isotope effect ($\alpha > 0$). However, as the anharmonic coefficient β increases (black and orange curves), the trend reverses, and T_c increases with increasing M , indicating the occurrence of the inverse isotope effect ($\alpha < 0$).

3 Conclusion

In this article, I have provided a brief interpretation of the inverse isotope effect observed in certain superconducting materials, such as palladium hydrides. Starting from the Fröhlich Hamiltonian, I approximately calculated the electron-phonon coupling constant λ based on the its corresponding in the BCS theory. Then, by introducing a one-dimensional anharmonic potential toy model and using the ground state wavefunction to calculate the zero-point motion, I demonstrated how anharmonic

effects modify the phonon frequencies and, consequently, the electron-phonon coupling constant λ . Finally, using the McMillan equation, I analyzed the superconducting critical temperature T_c and showed that anharmonic effects can lead to an inverse isotope effect, where T_c increases with increasing ionic mass.

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